

C# Console Application Assignment

Word Guessing Game

Build a console-based word guessing game similar to Wordle using **C#**, **Object-Oriented Programming (OOP)**, and **Exception Handling**.

Game Rules

- The system should secretly choose one 5-letter word.
- The player gets a maximum of 6 attempts.
- The user should enter one guess per attempt.
- If the word is guessed correctly, the game should end immediately.

Example Hidden Words

- APPLE
- MANGO
- GRAPE
- TRAIN
- PLANT
- BRAIN

Feedback Rules

- G = Correct letter in correct position
- Y = Correct letter in wrong position
- X = Letter not present in the word

Sample Feedback

Hidden Word: MANGO

User Guess: MAGIC

Output:

M A G I C

G G Y X X

Attempt-Based Comments

Attempt	Comment
1	Genius!
2	Excellent!
3	Great job!
4	Good work!
5	Nice try!
6	That was close!

OOP Requirements

- **Game** class to manage game flow and attempts.
- **WordProvider** class to provide random words.
- **GuessValidator** class to validate user input.
- **FeedbackGenerator** class to generate G/Y/X feedback.
- **InvalidGuessException** custom exception class.

Exception Handling Requirements

- Empty input
- Input less than 5 letters
- Input greater than 5 letters
- Input containing numbers
- Input containing special characters

Sample Invalid Inputs

CAR
BANANA
AB12E
@PPLE

Bonus Requirements

- Replay option
- Color output using `Console.ForegroundColor`
- Difficulty levels
- Score calculation
- Prevent duplicate guesses

Expected Concepts Used

- Classes and Objects
- Encapsulation
- Constructors
- Methods
- Collections / Lists
- Loops
- Conditional Statements
- Custom Exceptions
- String Handling

Submission Instructions:

Submit the complete Visual Studio solution folder along with screenshots of successful execution.